

E-commerce Data Quality Project Documentation

1. Dataset structure

The project contains three related datasets. The processed files keep the same columns as the raw files.

Table	Primary Key	Foreign Keys
customers	customer_id	None
products	product_id	None
orders	order_id	customer_id -> customers.customer_id; product_id -> products.product_id

2. Profiling summary

Dataset	Raw rows	Processed rows	Verified change
Customers	510	500	10 duplicated customer IDs were identified and reduced to one record per customer ID; 3 blank customer IDs were completed using the missing values in the customer ID sequence as a documented cleaning assumption.
Products	52	50	2 duplicate product rows removed
Orders	2,030	1,947	30 duplicate order rows removed and 53 invalid orders excluded

3. Issues found and handling

Customers

Issue found in raw data	Handling visible in processed data
10 duplicated customer IDs (20 rows)	Duplicates were reduced to one row per customer ID.
3 customer records had blank customer_id	Assigned CUST-00222, CUST-00249 and CUST-00280 using the missing values in the customer ID sequence. This was treated as a project cleaning assumption because the source rows themselves did not contain the IDs.
Whitespace and inconsistent name/country/email capitalization	Text was trimmed/standardised; known UK country variants were standardised to United Kingdom and emails were

	lowercased.
Mixed signup-date formats	Valid dates were converted to YYYY-MM-DD.
2 impossible signup dates	They are blank/NULL in the processed data.
Missing email values	They remain missing because email was treated as optional.

Products

Issue found in raw data	Handling visible in processed data
2 duplicated product IDs: PROD-0001 and PROD-0044	Exact duplicate rows were reduced to one row per product ID.
2 missing categories: PROD-0036 and PROD-0042	Filled as Books and Toys & Games using the matching brand/category evidence used during cleaning.
Invalid prices: PROD-0032 = -14.99, PROD-0038 = -34.99, PROD-0048 = 0.00	Recovered as 14.99, 34.99 and 9.49 from consistent positive prices in order data.
Minor category whitespace	Trimmed in the processed data.

Orders

Issue found in raw data	Handling visible in processed data
30 duplicated order IDs (60 rows)	One record per duplicated order ID was retained; no duplicated order ID was removed completely.
8 blank customer IDs and 10 customer IDs not found in raw customers	These orders were excluded under the adopted rule that every clean order needs a valid customer.
6 product IDs not found in the product table	These orders were excluded under the adopted rule that every clean order needs a valid product.
4 impossible order dates	These orders were excluded because order_date is mandatory.
5 missing quantities and 5 zero quantities	These orders were excluded because quantity must be present and greater than zero.
6 quantities of -1	Corrected to 1 in the processed data following the cleaning decision made during the project.
3 negative order prices	Corrected to positive values 74.99, 22.99 and 16.99 using the corresponding product/order evidence.
15 missing payment methods	These orders were excluded because payment_method is mandatory under the adopted rules.
Inconsistent payment/status capitalization and Delivred typo	Formatting was standardised and Delivred was corrected to Delivered.

Mixed valid date formats	Converted to YYYY-MM-DD.
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4. Processed-data validation

The processed files were checked again after cleaning. These checks passed:

- Customer, product and order IDs were checked for uniqueness in the processed data.
- Customers, products and orders were checked for duplicate IDs, and no duplicate primary keys remain.
- Products were checked to make sure required fields were present and prices were greater than zero.
- Orders were checked to make sure required fields were present, quantity was greater than zero and unit price was greater than zero.
- Every customer_id in orders exists in the processed customers table, and every product_id in orders exists in the processed products table.
- The final processed files contain 500 customers, 50 products and 1,947 valid orders.

5. Cleaning assumptions and decisions

Three raw customer records had blank customer IDs. During cleaning, CUST-00222, CUST-00249 and CUST-00280 were assigned because these were the missing values in the otherwise sequential customer ID range. This was a documented project assumption used to retain those customer records rather than discard them.

The sequence-based customer ID assignment is therefore recorded as an assumption rather than as a value directly recovered from the raw source.

6. Short process write-up

The original data was kept in the raw schema and the cleaned data was created in a separate clean schema. Checked the three tables for duplicate records, missing values, inconsistent text, invalid dates, invalid prices or quantities, and incorrect customer or product references. Duplicates were removed, text and dates were standardised, and values were corrected when there was clear evidence. Orders that were still missing required information were removed from the clean orders table. Finally, we checked that IDs were unique, product prices and order quantities were valid, and every order linked to an existing customer and product. For the three customer records with missing IDs, the missing numbers in the customer ID sequence were used as a documented assumption.